Project Presentation of CO-OP Project at Industry (Module-2) (22CS421)

On

Low Power Edge Inference for Green Computing: An Evaluation of Post-Training Quantization on Apple Silicon

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Introduction

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The Challenge

Large language models (LLMs) are transforming NLP — but at significant energy cost. The IEA projects data centers will consume 945 TWh/year by 2030, with AI inference accounting for ~60% of ML compute in production.

Energy

Inference consumes ~60% of total ML compute in production systems

Edge Constraints

Edge devices limited to 8–16 GB RAM, tight power budgets

Model Size

A 4B-param FP16 model needs ~8 GB for weights alone

Solution: Post-Training Quantization (PTQ)

PTQ reduces weight precision from FP16/FP32 to INT8 or INT4 without retraining — enabling existing models to fit on resource-constrained edge hardware. Apple Silicon's unified memory architecture amplifies these benefits.

3.5×–3.6×
Memory reduction with INT4

2.0×–3.8×
Throughput boost with INT4

< 2.5 GB
All models at INT4 precision

Research Objectives & Motivation

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1. Evaluate how quantization bit-widths (FP16, INT8, INT4) affect throughput, memory, and accuracy on Apple Silicon.
2. Determine how model size influences quantization sensitivity and the accuracy-efficiency trade-off.
3. Assess Apple's MLX framework as a platform for efficient edge inference.
4. Provide actionable recommendations for practitioners deploying SLMs on edge hardware.

Methodology & Experimental Setup

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Models Evaluated

Qwen3.5-0.8B

873M params — extreme efficiency

Qwen3.5-2B

2.27B params — balanced performance

Qwen3.5-4B

4.66B params — high accuracy

Phi-3 Mini

3.82B params — dense transformer (Microsoft)

Hardware & Protocol

Platform:	Apple MacBook Pro — M4 Pro Chip
CPU / GPU:	12-core CPU, 16-core GPU
Memory:	24 GB LPDDR5X Unified Memory
Framework:	Apple MLX 0.31.0
Generation:	100 tokens, 256-token input context
Accuracy:	Full MMLU — 57 subjects, 14,042 Qs
Decoding:	Greedy (temp = 0.0), seed = 42

Tools & Technologies

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Apple MLX 0.31.0

Native Apple Silicon inference framework with built-in FP16/INT8/INT4 quantization

Python 3.x

Primary scripting language for benchmarking, evaluation, and automation

Hugging Face Transformers

Model download, tokenizer access, and evaluation utilities

Custom MMLU Evaluator

Logit-based next-token scoring over A/B/C/D across 14,042 questions

CodeCarbon 3.2.3

Estimated energy/CO_2 tracking during inference (TDP-based on Apple Silicon)

lm-evaluation-harness

Reference methodology for logit-based multiple-choice evaluation

Implementation

Implementation Workflow

1 Model Loading — Models downloaded from HuggingFace Hub, loaded via mlx-lm

2 Quantization — PTQ applied: FP16 → INT8 / INT4 using MLX built-in utilities. No retraining needed.

3 Throughput Benchmark — 100 tokens generated, 256-token context. 10 runs after 1 warmup. Mean tok/s recorded.

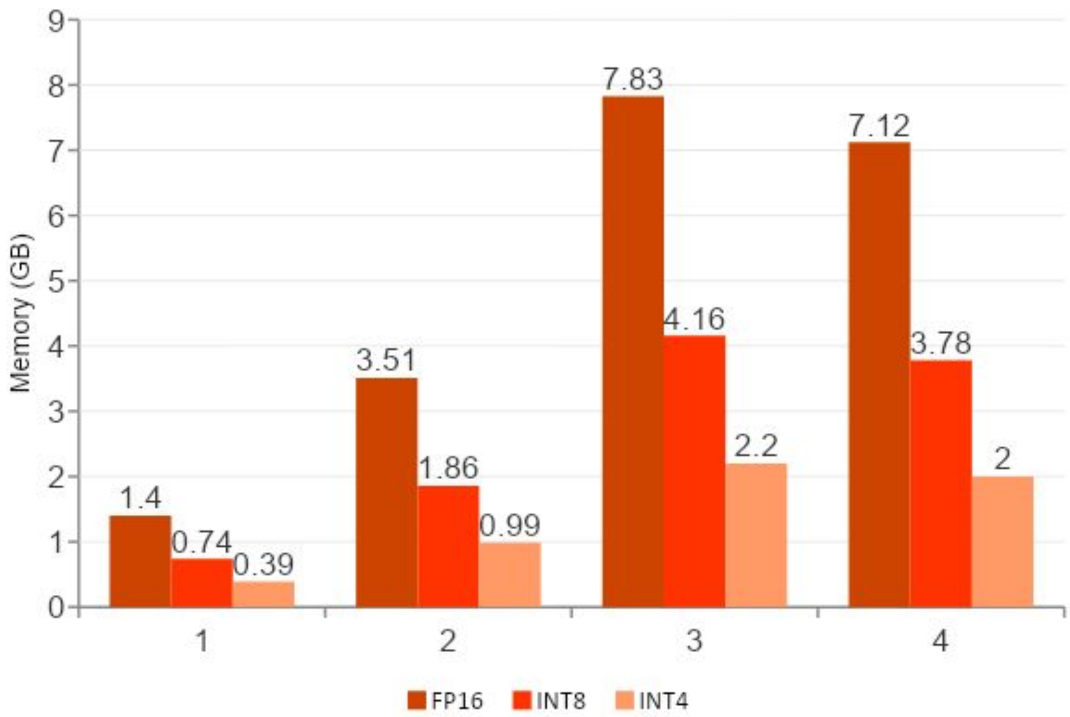
4 MMELU Accuracy — Logit-based scoring across 57 subjects (14,042 Qs). Next-token log-probs over A/B/C/D.

5 Energy Tracking — CodeCarbon wraps each inference pass. kWh estimated per run, averaged over 10 runs.

Files can be found here! https://github.com/harshpreet931/Low-Power-Edge-Inference-for-Green-Computing

Results: Memory Footprint

INT4 quantization delivers 3.5×–3.6× memory reduction, every model fits within 2.5 GB, enabling 8 GB edge deployment.



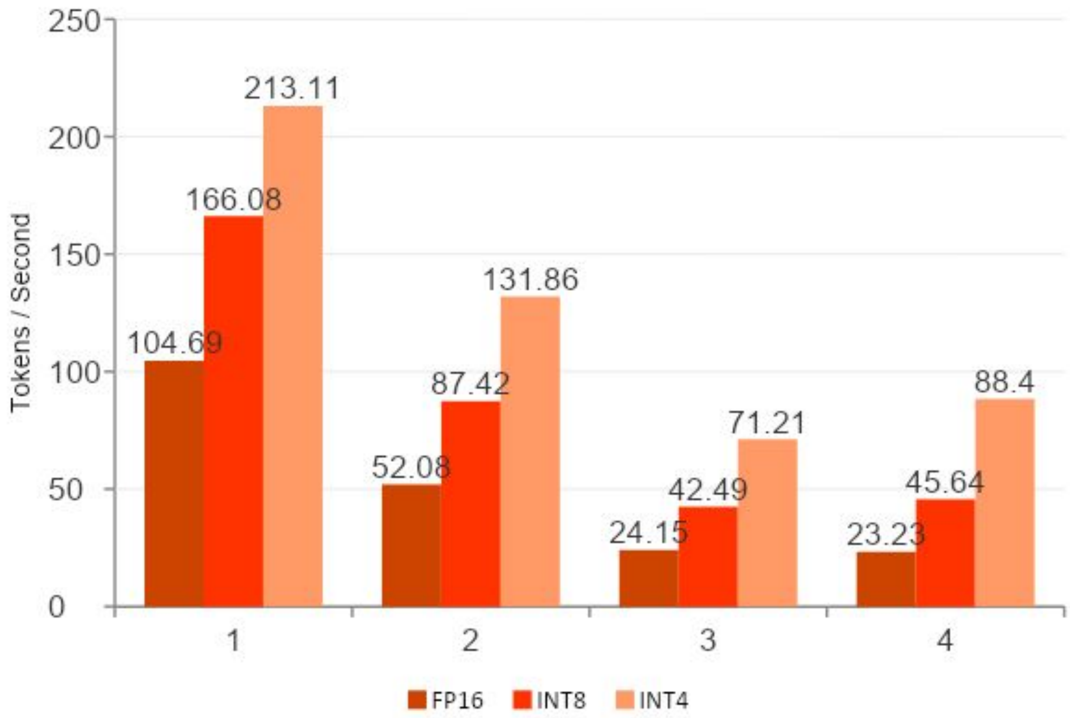
2.5 GB
Max memory at INT4 (all models)

3.6×
Memory reduction (Qwen3.5-4B: 7.83→2.20 GB)

< 1 GB
Qwen3.5-0.8B at INT4

Results: Throughput Analysis

INT4 speedup increases with model size, larger models are more memory-bandwidth-bound, so bandwidth reduction helps proportionally more.



2.0×
Qwen3.5-0.8B INT4/FP16

2.5×
Qwen3.5-2B INT4/FP16

2.9×
Qwen3.5-4B INT4/FP16

3.8×
Phi-3 Mini INT4/FP16 (best)

Results: MMLU Accuracy

Full MMLU (14,042 Qs), ±0.8 pp confidence interval. INT8 is effectively lossless for 2B+ models. INT4 drops are real but model-dependent.

Model	FP16 %	INT8 %	INT4 %	INT4 Drop
Qwen3.5-0.8B	46.31	45.27	43.73	**−2.59 pp**
Qwen3.5-2B	56.59	56.54	48.68	**−7.90 pp**
Qwen3.5-4B	73.11	73.07	69.82	**−3.29 pp**
Phi-3 Mini	69.36	69.29	66.97	**−2.39 pp**

Key Insights

- **INT8 near-lossless** for 2B+ models (< 0.1 pp drop)
- **INT4 sensitivity is architecture-dependent**, not just parameter count
- **4B-class models absorb INT4 best** (2.39–3.29 pp drop only)

Best Trade-offs

Phi-3 Mini INT4

2.39 pp drop
73.7% energy savings

Qwen3.5-4B INT4

3.29 pp drop
66.1% energy savings

Any 2B+ INT8

< 0.1 pp drop
~45% energy savings

Conclusion & Future Scope

Conclusion

- INT4 via MLX: 2.0×–3.8× throughput, 3.5×–3.6× memory reduction. All models fit < 2.5 GB.
- INT8 is effectively lossless for 2B+ models (< 0.1 pp MMLU drop) — the safe accuracy-first default.
- INT4 causes 2.39–7.90 pp MMLU drops — choose per model, not per bit-width alone.
- Phi-3 Mini INT4 offers the best energy-accuracy trade-off: 73.7% energy savings for 2.39 pp drop.
- Apple Silicon's unified memory uniquely amplifies quantization benefits vs. discrete-GPU systems.

Future Scope

→ Hardware Power Measurement:

External power meters or Apple powermetrics for ground-truth energy data

→ Broader Benchmarks:

Reasoning (GSM8K), coding (HumanEval), long-context (SCROLLS) beyond MMLU

→ Cross-Architecture:

NVIDIA GPUs, mobile NPUs, Raspberry Pi — generalize recommendations

→ Dynamic Quantization:

Per-layer adaptive bit-width based on sensitivity analysis

→ Combined Techniques:

Quantization + pruning + distillation for deeper compression

Thank You